MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, EXCLUSIVE OR, DIGITAL, KEY CODE EBB

INACTIVE FOR DESIGN AFTER
DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic exclusive OR gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.6 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/171-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

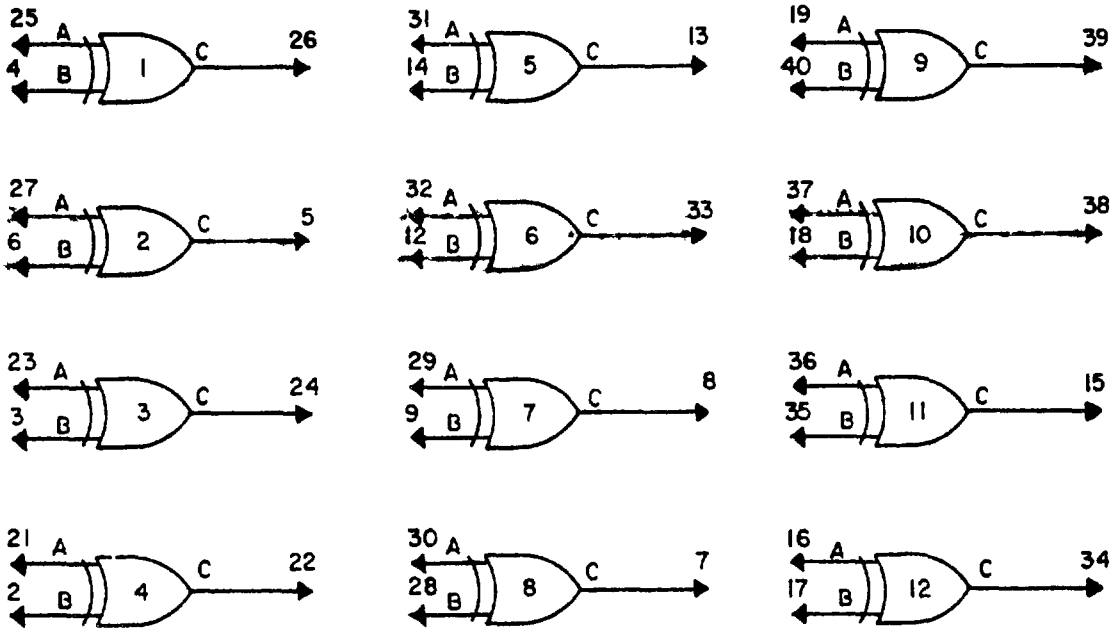
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

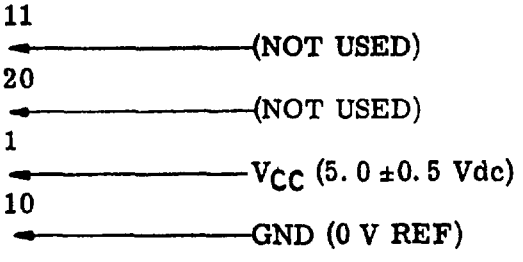
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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* FIGURE 1. EBB module function diagram, exclusive OR gate.

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TABLE I. Module electrical requirements.

	Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirement 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
					Initial		End-of-life		Initial		End-of-life		
					Min	Max	Min	Max	Min	Max	Min	Max	
	A1-A12	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
	B1-B24	Input breakdown current	IV	S	0	200	0	200	0	200	0	200	µA
	C1	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
	C2, C3	Isolation, unused pins	V	X	10	∞	1	∞	10	∞	1	∞	MΩ
	D1-D24	Static output voltage, low level	VI	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
	E1-E24	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
*	F1-F24	Input current, low level	VIII	X	50	400	50	400	50	400	50	400	µA
	G1-G24	Input current, high level	IX	X	0	20	0	20	0	20	0	20	µA
	H1-H12	Output short circuit current	X	X	3	15	3	15	3	15	3	15	mA
	J1, J2	Power supply current, low level	XI	S	2	20	2	20	2	20	2	20	mA
*	J3, J4	Power supply current, high level	XI	S	1.3	13.0	1.3	13.0	1.3	13.0	1.3	13.0	mA
	K1-K12	Transition time (t_{TLH}), C_L = 40 pF	XII	S	5	60	5	60	5	60	5	60	ns
	K13-K24	Transition time (t_{TLH}), C_L = 140 pF	XII	Q	10	100	10	100	10	100	10	100	ns
	L1-L12	Transition time (t_{THL}), C_L = 40 pF	XIII	S	2	30	2	30	2	30	2	30	ns
	L13-L24	Transition time (t_{THL}), C_L = 140 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
	M1-M24	Propagation delay time (t_{PLH}), other input high, C_L = 40 pF	XIV	S	10	90	10	90	10	110	10	110	ns
	M25-M48	Propagation delay time (t_{PLH}), other input low, C_L = 40 pF	XIV	S	10	150	10	150	10	165	10	165	ns
	M49-M72	Propagation delay time (t_{PLH}), other input high, C_L = 140 pF	XIV	Q	10	120	10	120	10	140	10	140	ns
	M73-M96	Propagation delay time (t_{PLH}), other input low, C_L = 140 pF	XIV	Q	10	180	10	180	10	195	10	195	ns
	N1-N24	Propagation delay time (t_{PHL}), other input high, C_L = 40 pF	XV	S	10	66	10	66	10	80	10	80	ns
	N25-N48	Propagation delay time (t_{PHL}), other input low, C_L = 40 pF	XV	S	10	150	10	150	10	165	10	165	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirement 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N49-N72	Propagation delay time (t_{PHL}), other input high, C_L = 140 pF	XV	Q	10	96	10	96	10	110	10	110	ns
N73-N96	Propagation delay time (t_{PHL}), other input low, C_L = 140 pF	XV	Q	10	180	10	180	10	195	10	195	ns
P1-P12	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
R1-R12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	300	0	300	0	300	0	300	mV
T	Power supply filter capacitance	XIX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
U1-U24	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through T shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/171-1 supersedes part number M28787/171. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
All outputs at low level	110 mW (outputs not loaded)
All outputs at high level	72.5 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 minimum, V_{CC} maximum

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Input current		
Low level	(V_{IL} = 0.0 V)	-400 µA maximum
	(V_{IL} = 0.3 V)	-360 µA maximum
High level	(V_{IH} = 2.4 V)	20 µA maximum
	(V_{IH} = 5.5 V)	200 µA maximum
Output voltage		
Low level	(I_{OL} = 2.0 mA)	0.3 V maximum
High level	(I_{OH} = -100 µA)	2.4 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.3 V)	2.0 mA
High level	(V_{OH} = 2.4 V)	-100 µA

* 6.5 Module function. This module contains 12 independent low-power exclusive OR gates.

6.5.1 Input to output relationship. Each low-power exclusive OR gate has the input to output relationship of table II for positve logic.

TABLE II. Input to output relationship.

Inputs		Outputs
A	B	Q
L	L	L
L	H	H
H	L	H
H	H	L



Notes: 1. H = high level.
2. L = low level.

6.5.2 Paralleling of outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

The circuit groups which can be paralleled are:

(1) Circuits 1, 2, 3, and 4.
(2) Circuits 5, 6, 7, and 8.
(3) Circuits 9, 10, 11, and 12.

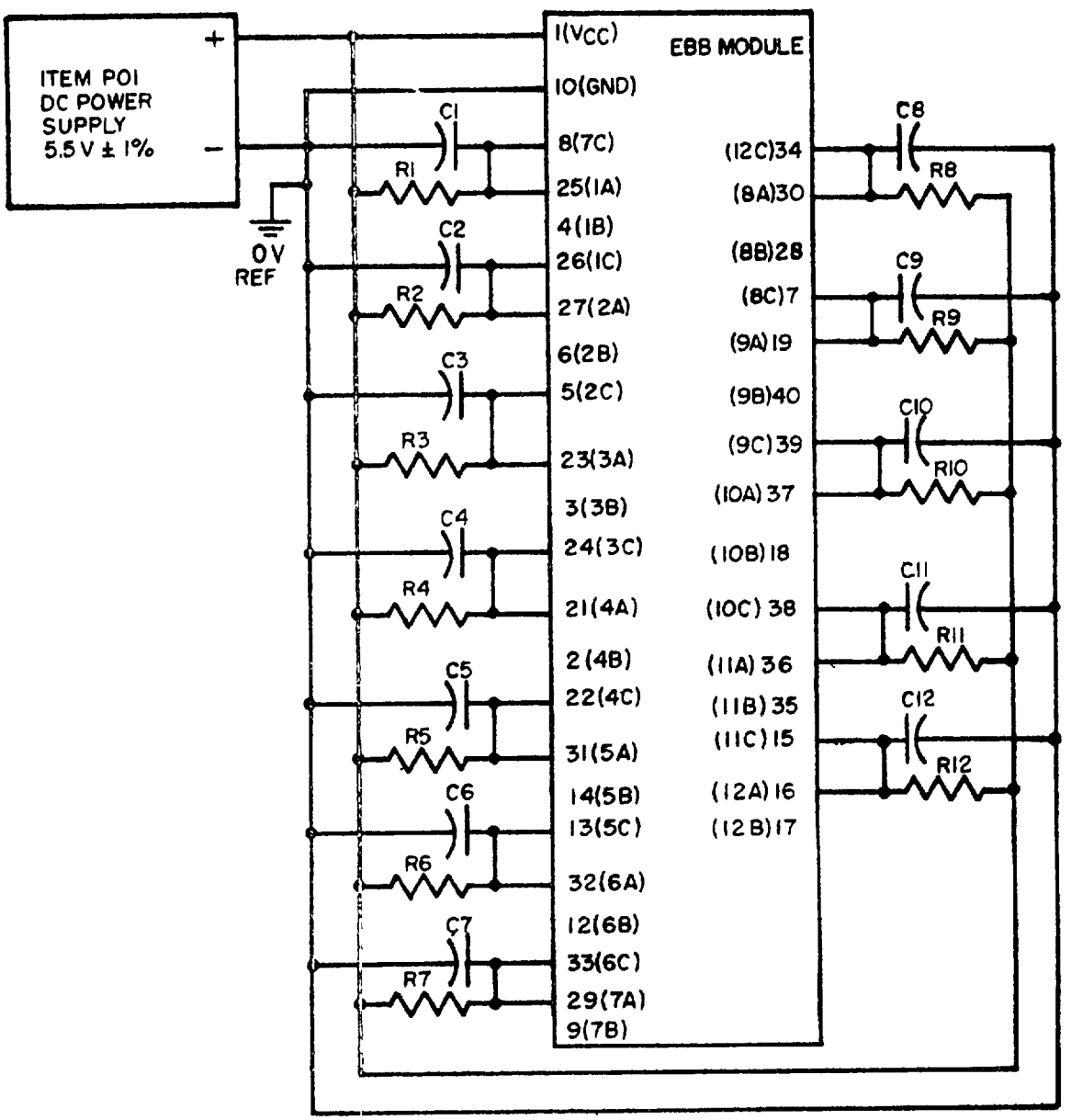
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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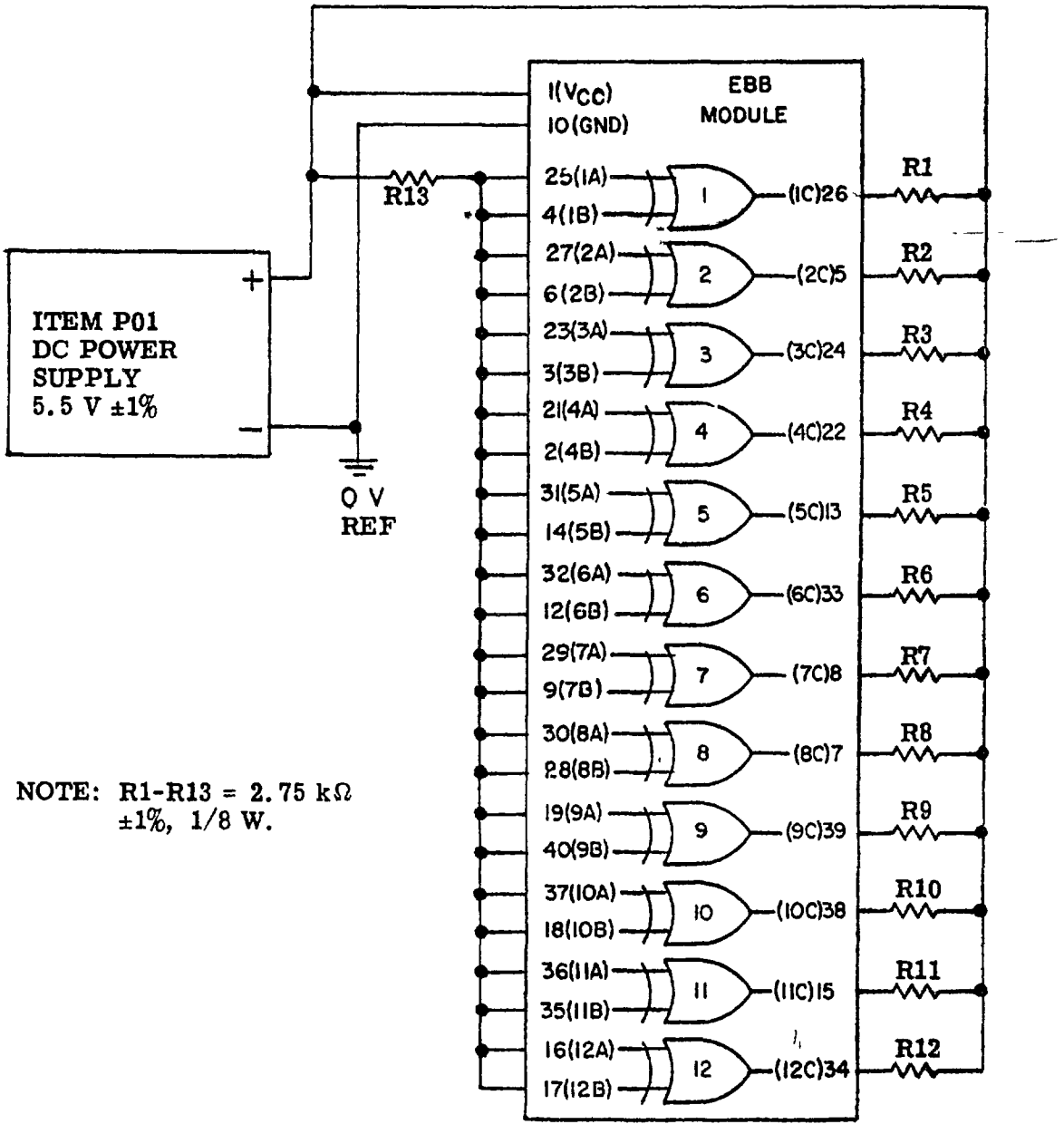
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NOTES:
1. R1-R12 = 2.75 kΩ ±1%, 1/8 W.
2. C1-C12 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R13 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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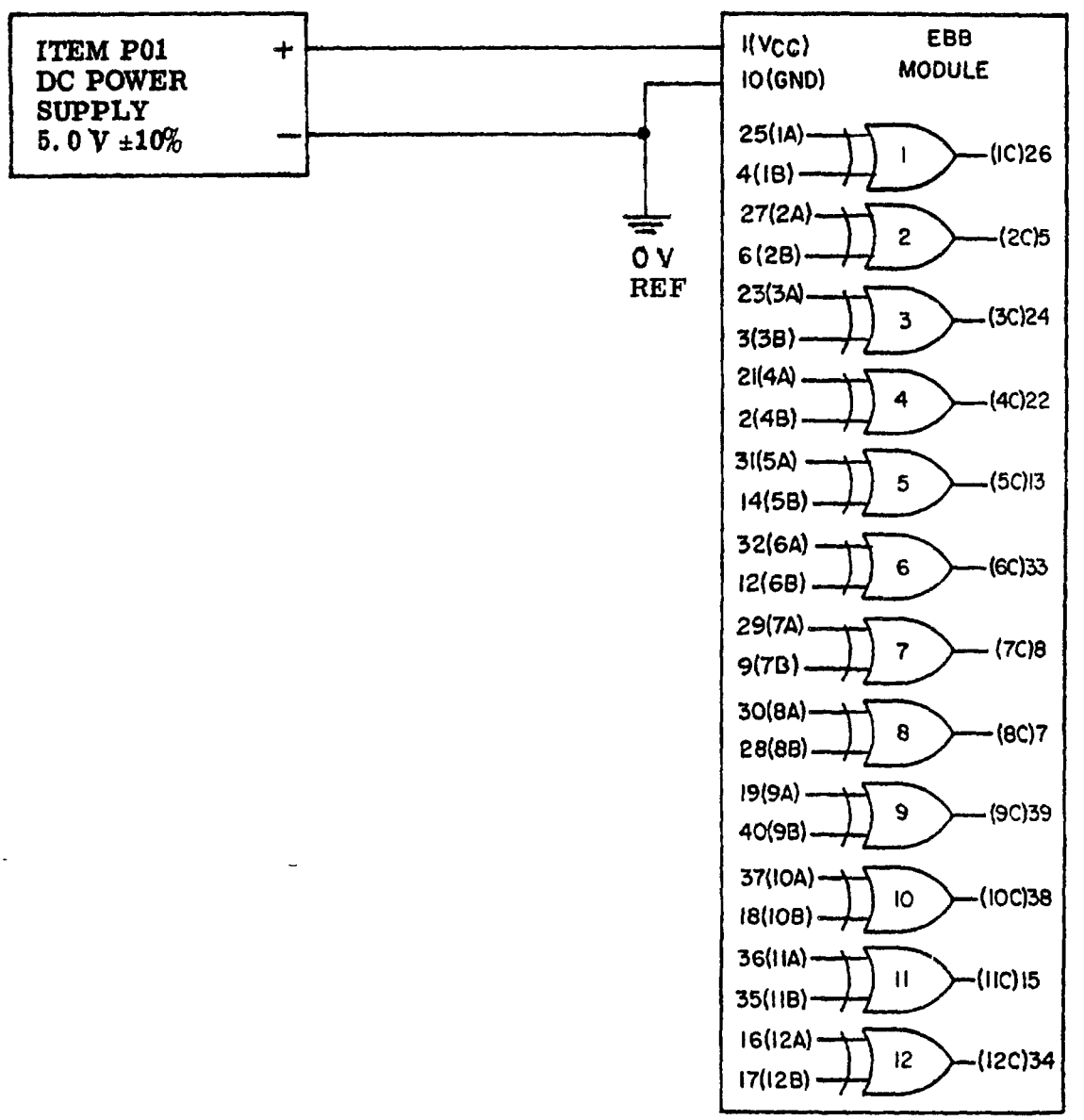


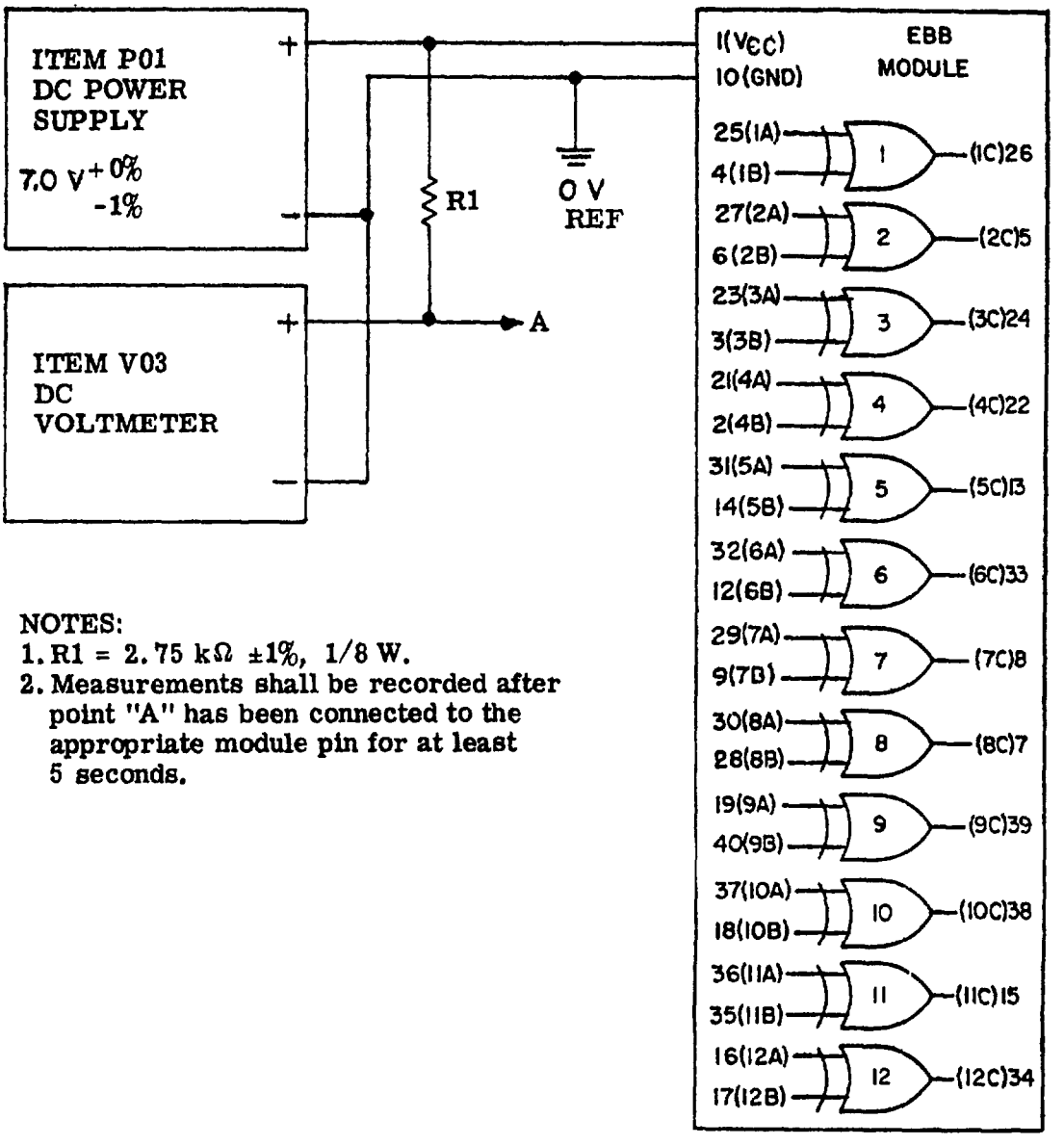
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1C V_{OUT} (low-loaded)	26	0.0	0.6	V	5
A2	2C	5	↓	↓	↓	↓
A3	3C	24				
A4	4C	22				
A5	5C	13				
A6	6C	33				
A7	7C	8				
A8	8C	7				
A9	9C	39				
A10	10C	38				
A11	11C	15				
A12	12C	34				

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point "A" has been connected to the appropriate module pin for at least 5 seconds.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Minimum	Maximum		
B1	1A Input breakdown current	25	4	0	200	µA	6
B2	1B	4	25				
B3	2A	27	6				
B4	2B	6	27				
B5	3A	23	3				
B6	3B	3	23				
B7	4A	21	2				
B8	4B	2	21				
B9	5A	31	14				
B10	5B	14	31				
B11	6A	32	12				
B12	6B	12	32				
B13	7A	29	9				
B14	7B	9	29				
B15	8A	30	28				
B16	8B	28	30				
B17	9A	19	40				
B18	9B	40	19				
B19	10A	37	18				
B20	10B	18	37				
B21	11A	36	35				
B22	11B	35	36				
B23	12A	16	17				
B24	12B	17	16				

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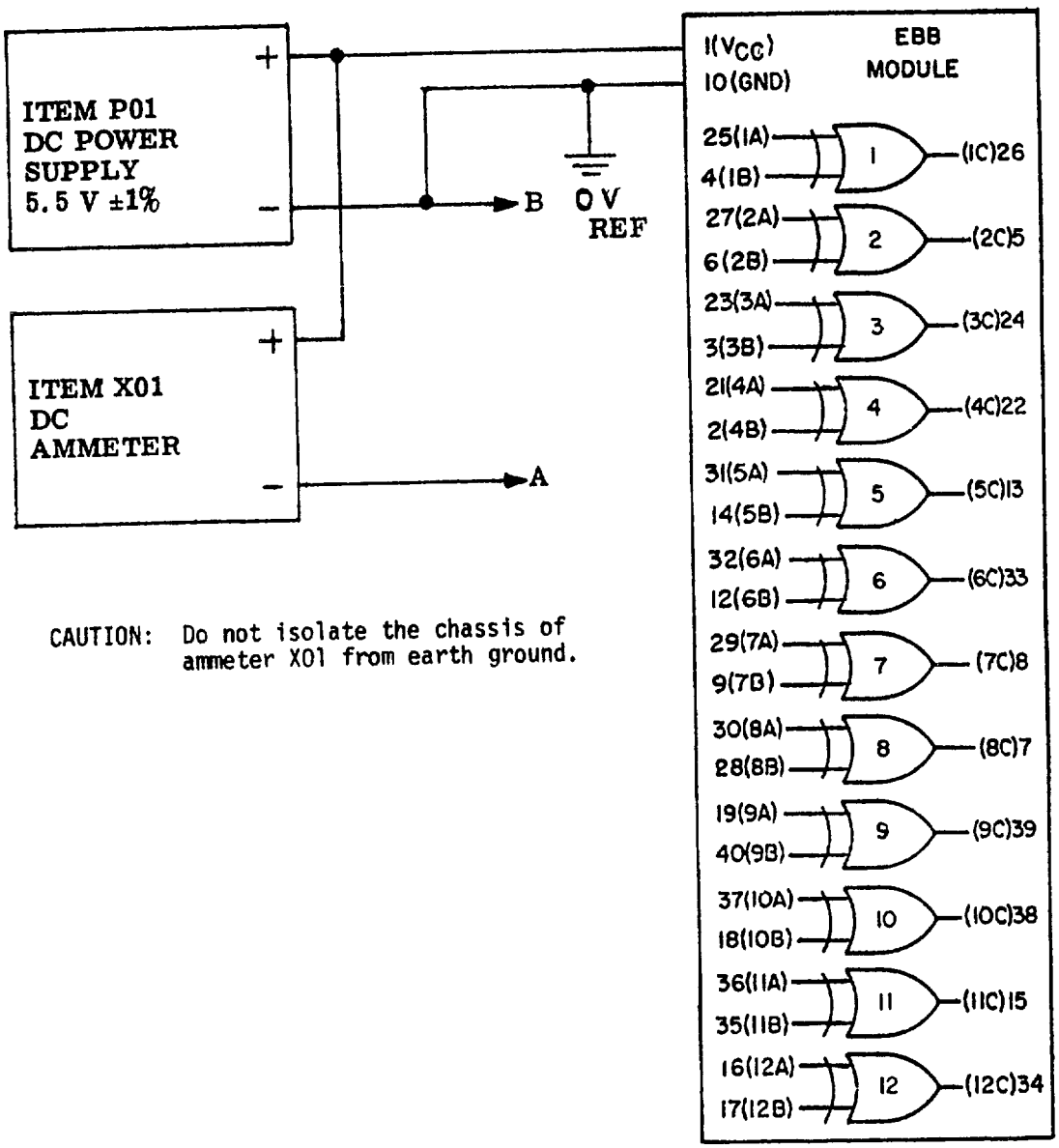


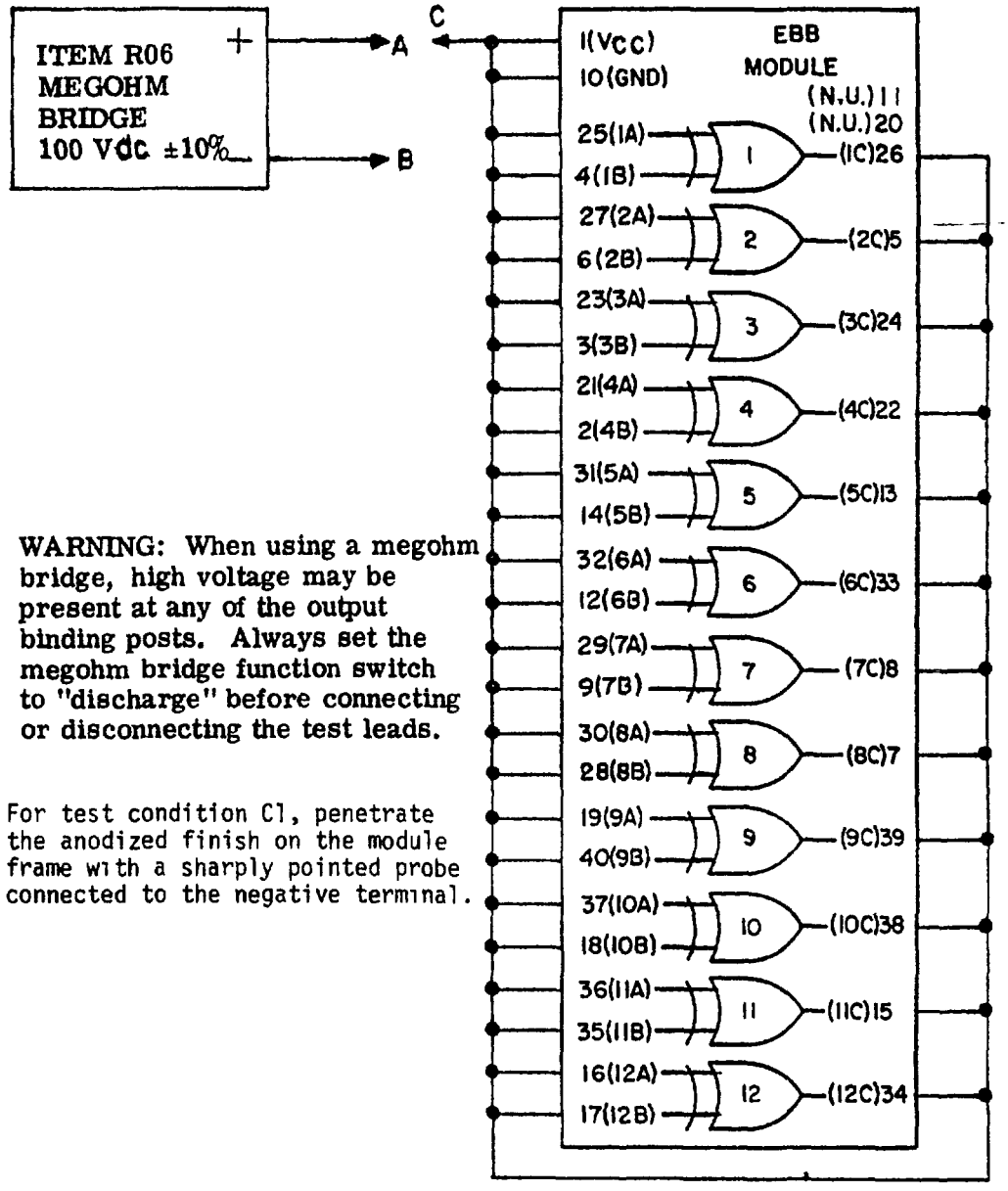
FIGURE 6. Input breakdown current test circuit for table IV.

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* TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame to module pins	C, 11, 20	Frame	10	∞	MΩ	7
C2	Isolation, used to unused pins	A	11, 20	10	∞	MΩ	7
C3	Isolation, Unused to unused pins	11	20	10	∞	MΩ	7

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WARNING: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition C1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Static output voltage (low-level) test.

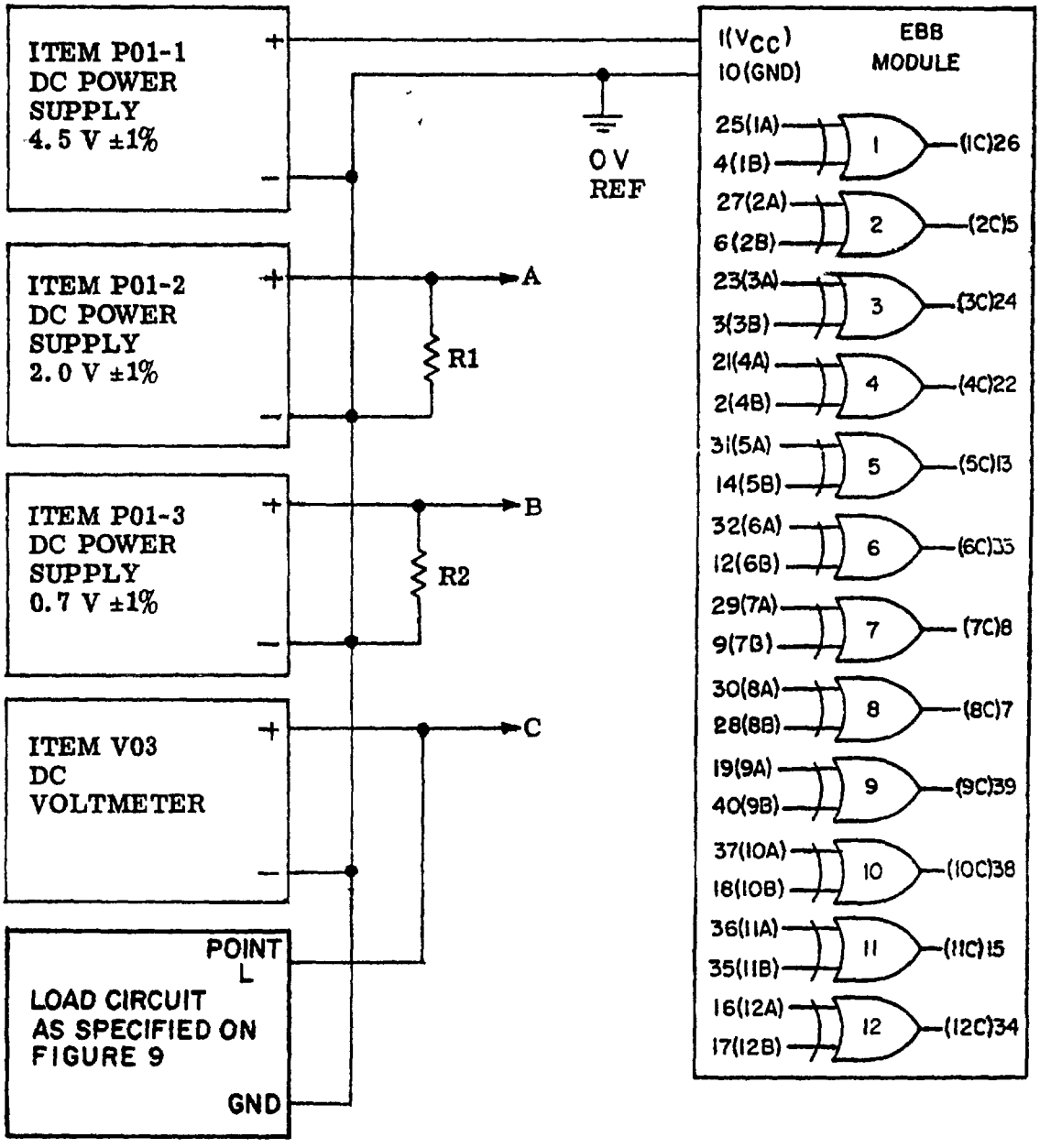
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1C Static output voltage, low level	25,4	Open	26	0.0	0.3	V	8
D2	2C	27,6		5				
D3	3C	23,3		24				
D4	4C	21,2		22				
D5	5C	31,14		13				
D6	6C	32,12		33				
D7	7C	29,9		8				
D8	8C	30,28		7				
D9	9C	19,40		39				
D10	10C	37,18		38				
D11	11C	36,35		15				
D12	12C	16,17		34				
D13	1C	Open	25,4	26				
D14	2C		27,6	5				
D15	3C		23,3	24				
D16	4C		21,2	22				
D17	5C		31,14	13				
D18	6C		32,12	33				
D19	7C		29,9	8				
D20	8C		30,28	7				
D21	9C		19,40	39				
D22	10C		37,18	38				
D23	11C		36,35	15				
D24	12C		16,17	34				

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TABLE VII. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1A Static output voltage, high level	25	4	26	2.4	4.5	V	8
E2	1B	4	25	26				
E3	2A	27	6	5				
E4	2B	6	27	5				
E5	3A	23	3	24				
E6	3B	3	23	24				
E7	4A	21	2	22				
E8	4B	2	21	22				
E9	5A	31	14	13				
E10	5B	14	31	13				
E11	6A	32	12	33				
E12	6B	12	32	33				
E13	7A	29	9	8				
E14	7B	9	29	8				
E15	8A	30	28	7				
E16	8B	28	30	7				
E17	9A	19	40	39				
E18	9B	40	19	39				
E19	10A	37	18	38				
E20	10B	18	37	38				
E21	11A	36	35	15				
E22	11B	35	36	15				
E23	12A	16	17	34				
E24	12B	17	16	34				

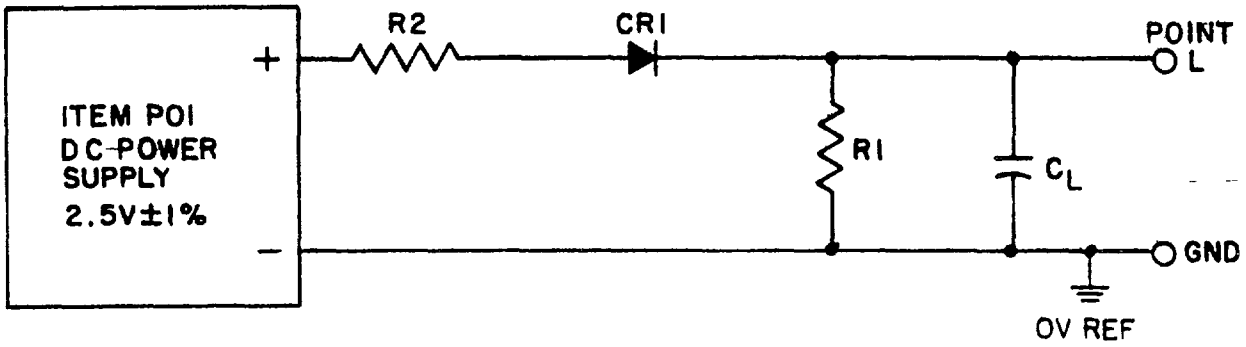
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NOTE: R1 = R2 = 50Ω ±5%, 1 W.

* FIGURE 8. Static output voltage (low-level and high-level) test circuit for tables VI, and VII.

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NOTES:
1. CR1 = 1N916, or equivalent.
2. R1 = 23.7 kΩ ±1%, 1/8 W with current flow of 100 µA out of point L when point L is held at 2.4 V.
3. R2 = select value for current flow of 2.0 mA ±5% out of point L when point L is held at 0.3 V.
4. For test conditions K1-K12, L1-L12, M1-M48, and N1-N48, C_L = 40 ±5 pF (including probe capacitance).
   For test conditions K13-K24, L13-L24, M49-M96, and N49-N96, C_L = 140 ±5 pF (including probe capacitance).
   For test conditions P1-P12, and R1-R12, C_L = 50 ±5 pF (including probe capacitance).
5. C_L is not required for dc electrical tests D and E (tables VI and VII).
6. The maximum lead length from point "L" to the appropriate module pin shall be less than 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

* FIGURE 9. Logic load circuit.

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*

TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1A Input current, low level	Open	25	Open	4	50	400	µA	10
F2	1B		4		25				
F3	2A		27		6				
F4	2B		6		27				
F5	3A		23		3				
F6	3B		3		23				
F7	4A		21		2				
F8	4B		2		21				
F9	5A		31		14				
F10	5B		14		31				
F11	6A		32		12				
F12	6B		12		32				
F13	7A		29		9				
F14	7B		9		29				
F15	8A		30		28				
F16	8B		28		30				
F17	9A		19		40				
F18	9B		40		19				
F19	10A		37		18				
F20	10B		18		37				
F21	11A		36		35				
F22	11B		35		36				
F23	12A		16		17				
F24	12B		17		16				

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TABLE IX. Input current (high-level) test.

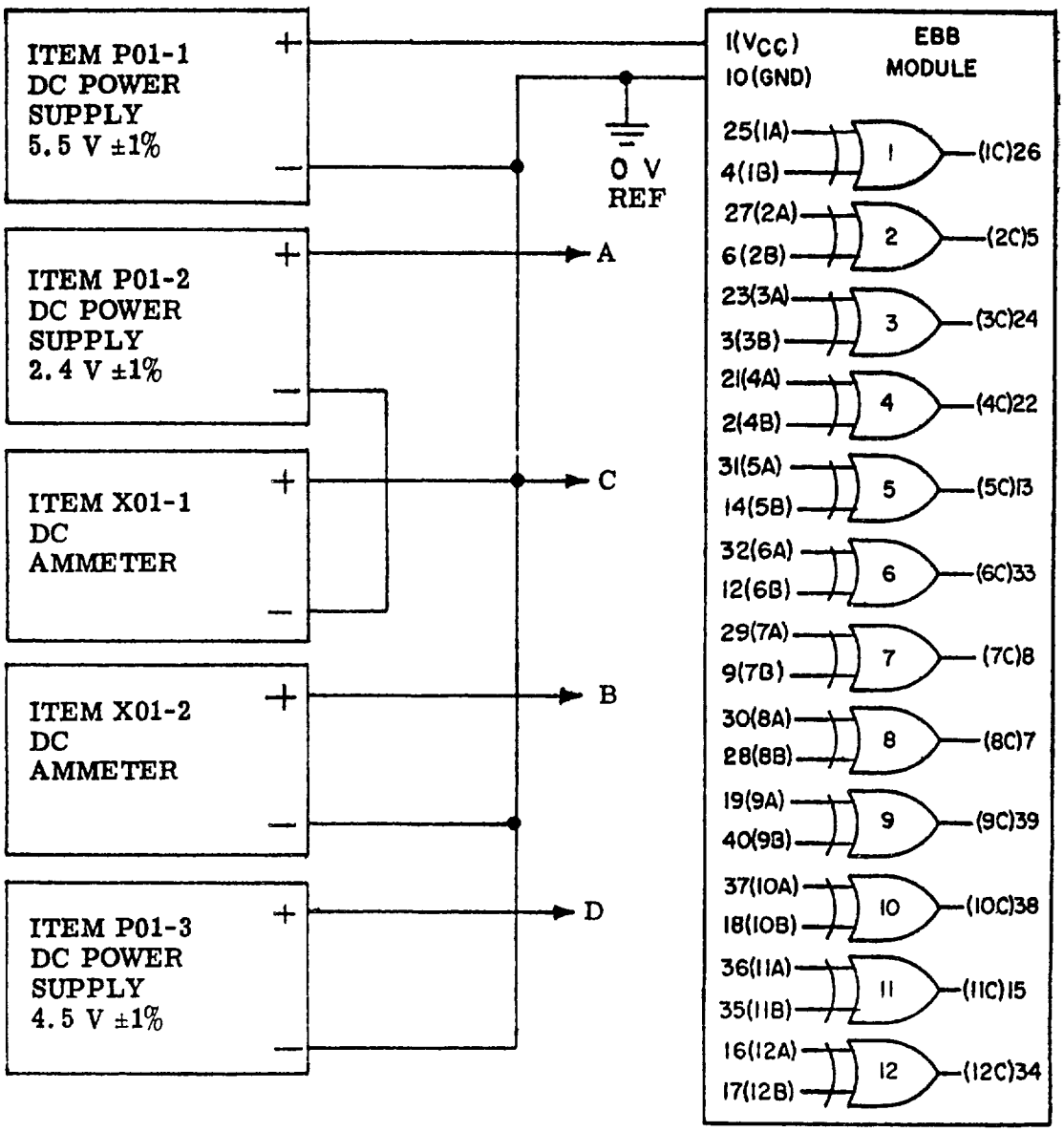
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1A Input current, high level	25	Open	4	Open	0	20	µA	10
G2	1B	4		25					
G3	2A	27		6					
G4	2B	6		27					
G5	3A	23		3					
G6	3B	3		23					
G7	4A	21		2					
G8	4B	2		21					
G9	5A	31		14					
G10	5B	14		31					
G11	6A	32		12					
G12	6B	12		32					
G13	7A	29		9					
G14	7B	9		29					
G15	8A	30		28					
G16	8B	28		30					
G17	9A	19		40					
G18	9B	40		19					
G19	10A	37		18					
G20	10B	18		37					
G21	11A	36		35					
G22	11B	35		36					
G23	12A	16		17					
G24	12B	17		16					

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1C Output short circuit current	25	26	4	Open	3	15	mA	10
H2	2C	27	5	6					
H3	3C	23	24	3					
H4	4C	21	22	2					
H5	5C	31	13	14					
H6	6C	32	33	12					
H7	7C	29	8	9					
H8	8C	30	7	28					
H9	9C	19	39	40					
H10	10C	37	38	18					
H11	11C	36	15	35					
H12	12C	16	34	17					

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. Since the ammeters are not used simultaneously, only one dc ammeter, X01, is required. Use ammeter X01-2 for tables VIII and X; use ammeter X01-1 for table IX.
2. Since the power supplies P01-2 and P01-3 are not used simultaneously, only two P01 supplies are required.
3. With point "A" disconnected, ammeter X01-1 should indicate zero current on the 1 μA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Power supply current, low level	C	C	A, B	Open	2	20	mA	11
J2	Power supply current, low level	D	D	Open	A, B	2	20	mA	11
J3	Power supply current, high level	C	D	A	B	1.3	13.0	mA	11
J4	Power supply current, high level	D	C	B	A	1.3	13.0	mA	11

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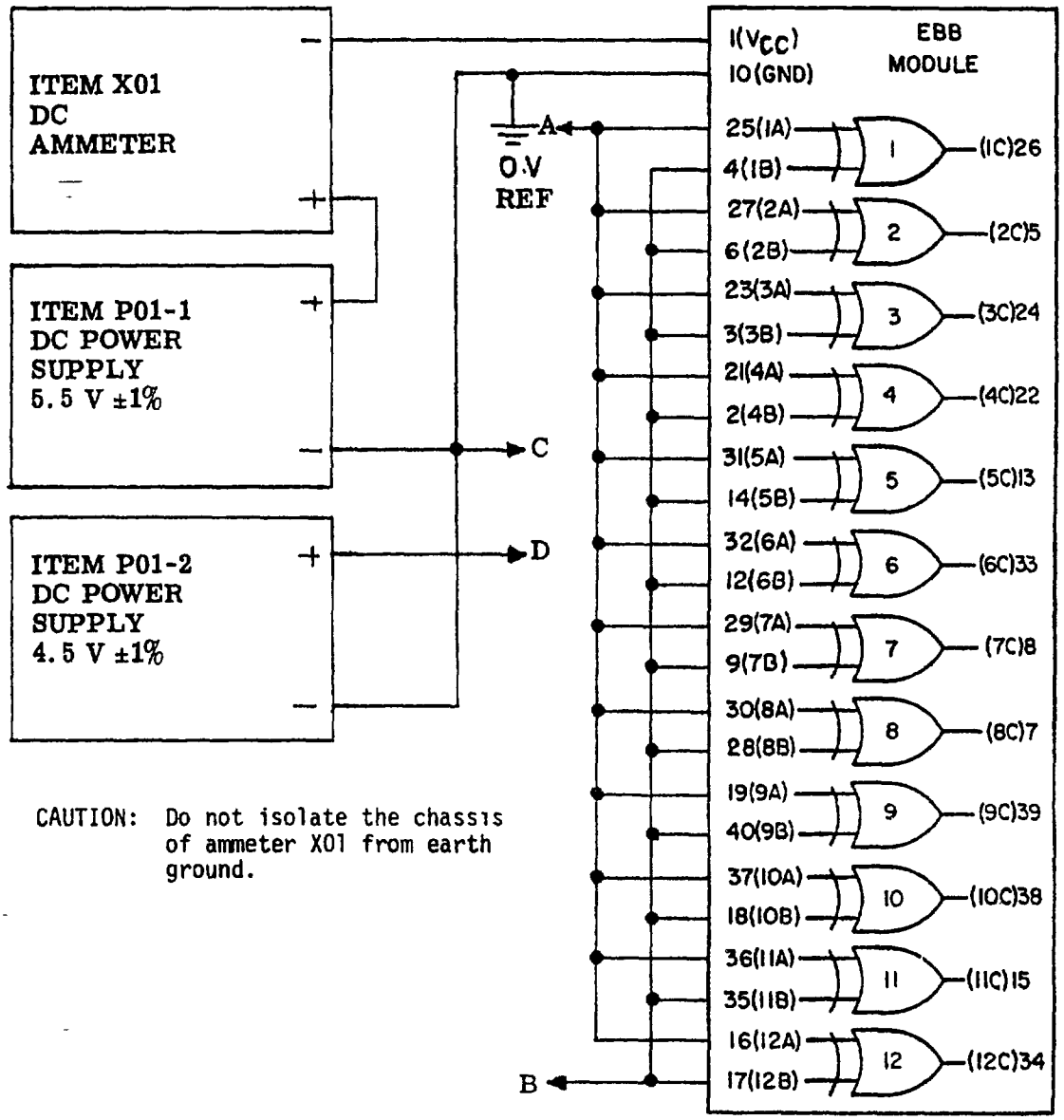


FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	1C Transition time (t_{TLH}), C_L = 40 pF	25	Open	4	26	5	60	ns	12, 13
K2	2C	27		6	5				
K3	3C	23		3	24				
K4	4C	21		2	22				
K5	5C	31		14	13				
K6	6C	32		12	33				
K7	7C	29		9	8				
K8	8C	30		28	7				
K9	9C	19		40	39				
K10	10C	37		18	38				
K11	11C	36		35	15				
K12	12C	16		17	34				
K13	1C Transition time (t_{TLH}), C_L = 140 pF	25		4	26	10	100		
K14	2C	27		6	5				
K15	3C	23		3	24				
K16	4C	21		2	22				
K17	5C	31		14	13				
K18	6C	32		12	33				
K19	7C	29		9	8				
K20	8C	30		28	7				
K21	9C	19		40	39				
K22	10C	37		18	38				
K23	11C	36		35	15				
K24	12C	16		17	34				

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1C Transition time (t_{THL}), C_L = 40 pF	25	Open	4	26	2	30	ns	12, 13
L2	2C	27		6	5				
L3	3C	23		3	24				
L4	4C	21		2	22				
L5	5C	31		14	13				
L6	6C	32		12	33				
L7	7C	29		9	8				
L8	8C	30		28	7				
L9	9C	19		40	39				
L10	10C	37		18	38				
L11	11C	36		35	15				
L12	12C	16		17	34				
L13	1C Transition time (t_{THL}), C_L = 140 pF	25		4	26	5	60		
L14	2C	27		6	5				
L15	3C	23		3	24				
L16	4C	21		2	22				
L17	5C	31		14	13				
L18	6C	32		12	33				
L19	7C	29		9	8				
L20	8C	30		28	7				
L21	9C	19		40	39				
L22	10C	37		18	38				
L23	11C	36		35	15				
L24	12C	16		17	34				

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TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1C Propagation delay time (t_{PLH}), other input high, C_L = 40 pF	25	4	Open	26	10	90	ns	12, 13
M2	2C	27	6		5				
M3	3C	23	3		24				
M4	4C	21	2		22				
M5	5C	31	14		13				
M6	6C	32	12		33				
M7	7C	29	9		8				
M8	8C	30	28		7				
M9	9C	19	40		39				
M10	10C	37	18		38				
M11	11C	36	35		15				
M12	12C	16	17		34				
M13	1C	4	25		26				
M14	2C	6	27		5				
M15	3C	3	23		24				
M16	4C	2	21		22				
M17	5C	14	31		13				
M18	6C	12	32		33				
M19	7C	9	29		8				
M20	8C	28	30		7				
M21	9C	40	19		39				
M22	10C	18	37		38				
M23	11C	35	36		15				
M24	12C	17	16		34				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M25	1C Propagation delay time (t_{PLH}), other input low, C_L = 40 pF	25	Open	4	26	10	150	ns	12, 13
M26	2C	27		6	5				
M27	3C	23		3	24				
M28	4C	21		2	22				
M29	5C	31		14	13				
M30	6C	32		12	33				
M31	7C	29		9	8				
M32	8C	30		28	7				
M33	9C	19		40	39				
M34	10C	37		18	38				
M35	11C	36		35	15				
M36	12C	16		17	34				
M37	1C	4		25	26				
M38	2C	6		27	5				
M39	3C	3		23	24				
M40	4C	2		21	22				
M41	5C	14		31	13				
M42	6C	12		32	33				
M43	7C	9		29	8				
M44	8C	28		30	7				
M45	9C	40		19	39				
M46	10C	18		37	38				
M47	11C	35		36	15				
M48	12C	17		16	34				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M49	1C Propagation delay time (t_{PLH}), other input high, C_L = 140 pF	25	4	Open	26	10	120	ns	12, 13
M50	2C	27	6		5				
M51	3C	23	3		24				
M52	4C	21	2		22				
M53	5C	31	14		13				
M54	6C	32	12		33				
M55	7C	29	9		8				
M56	8C	30	28		7				
M57	9C	19	40		39				
M58	10C	37	18		38				
M59	11C	36	35		15				
M60	12C	16	17		34				
M61	1C	4	25		26				
M62	2C	6	27		5				
M63	3C	3	23		24				
M64	4C	2	21		22				
M65	5C	14	31		13				
M66	6C	12	32		33				
M67	7C	9	29		8				
M68	8C	28	30		7				
M69	9C	40	19		39				
M70	10C	18	37		38				
M71	11C	35	36		15				
M72	12C	17	16		34				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M73	1C Propagation delay time (t_{PLH}), other input low, C_L = 140 pF	25	Open	4	26	10	180	ns	12, 13
M74	2C	27		6	5				
M75	3C	23		3	24				
M76	4C	21		2	22				
M77	5C	31		14	13				
M78	6C	32		12	33				
M79	7C	29		9	8				
M80	8C	30		28	7				
M81	9C	19		40	39				
M82	10C	37		18	38				
M83	11C	36		35	15				
M84	12C	16		17	34				
M85	1C	4		25	26				
M86	2C	6		27	5				
M87	3C	3		23	24				
M88	4C	2		21	22				
M89	5C	14		31	13				
M90	6C	12		32	33				
M91	7C	9		29	8				
M92	8C	28		30	7				
M93	9C	40		19	39				
M94	10C	18		37	38				
M95	11C	35		36	15				
M96	12C	17		16	34				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1C Propagation delay time (t_{PHL}), other input high, C_L = 40 pF	25	4	Open	26	10	66	ns	12, 13
N2	2C	27	6		5				
N3	3C	23	3		24				
N4	4C	21	2		22				
N5	5C	31	14		13				
N6	6C	32	12		33				
N7	7C	29	9		8				
N8	8C	30	28		7				
N9	9C	19	40		39				
N10	10C	37	18		38				
N11	11C	36	35		15				
N12	12C	16	17		34				
N13	1C	4	25		26				
N14	2C	6	27		5				
N15	3C	3	23		24				
N16	4C	2	21		22				
N17	5C	14	31		13				
N18	6C	12	32		33				
N19	7C	9	29		8				
N20	8C	28	30		7				
N21	9C	40	19		39				
N22	10C	18	37		38				
N23	11C	35	36		15				
N24	12C	17	16		34				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N25	1C Propagation delay time (t_{PHL}), other input low, C_L = 40 pF	25	Open	4	26	10	150	ns	12, 13
N26	2C	27		6	5				
N27	3C	23		3	24				
N28	4C	21		2	22				
N29	5C	31		14	13				
N30	6C	32		12	33				
N31	7C	29		9	8				
N32	8C	30		28	7				
N33	9C	19		40	39				
N34	10C	37		18	38				
N35	11C	36		35	15				
N36	12C	16		17	34				
N37	1C	4		25	26				
N38	2C	6		27	5				
N39	3C	3		23	24				
N40	4C	2		21	22				
N41	5C	14		31	13				
N42	6C	12		32	33				
N43	7C	9		29	8				
N44	8C	28		30	7				
N45	9C	40		19	39				
N46	10C	18		37	38				
N47	11C	35		36	15				
N48	12C	17		16	34				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

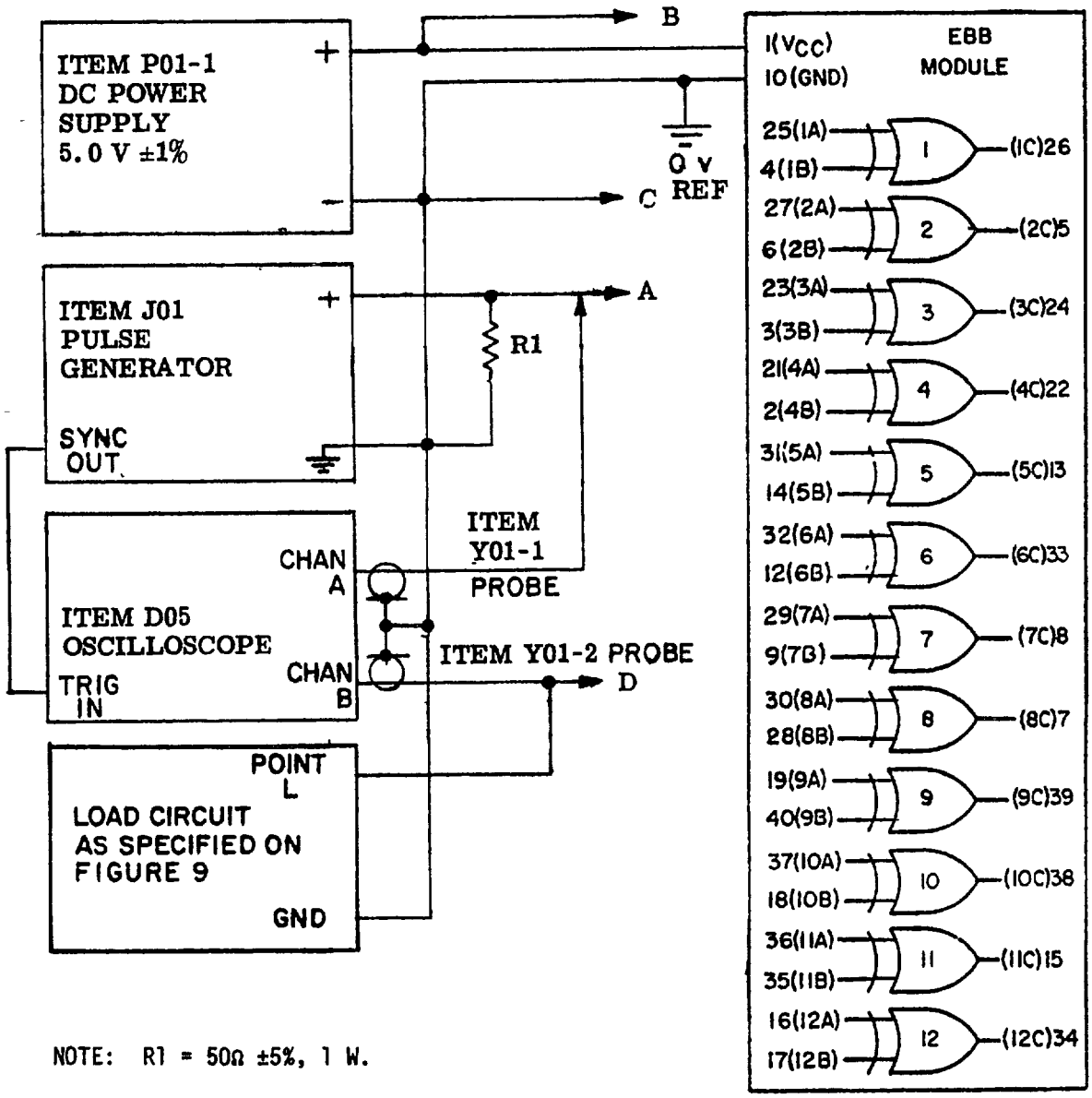
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N49	1C Propagation delay time (t_{PHL}), other input high, C_L = 140 pF	25	4	Open	26	10	96	ns	12, 13
N50	2C	27	6		5				
N51	3C	23	3		24				
N52	4C	21	2		22				
N53	5C	31	14		13				
N54	6C	32	12		33				
N55	7C	29	9		8				
N56	8C	30	28		7				
N57	9C	19	40		39				
N58	10C	37	18		38				
N59	11C	36	35		15				
N60	12C	16	17		34				
N61	1C	4	25		26				
N62	2C	6	27		5				
N63	3C	3	23		24				
N64	4C	2	21		22				
N65	5C	14	31		13				
N66	6C	12	32		33				
N67	7C	9	29		8				
N68	8C	28	30		7				
N69	9C	40	19		39				
N70	10C	18	37		38				
N71	11C	35	36		15				
N72	12C	17	16		34				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

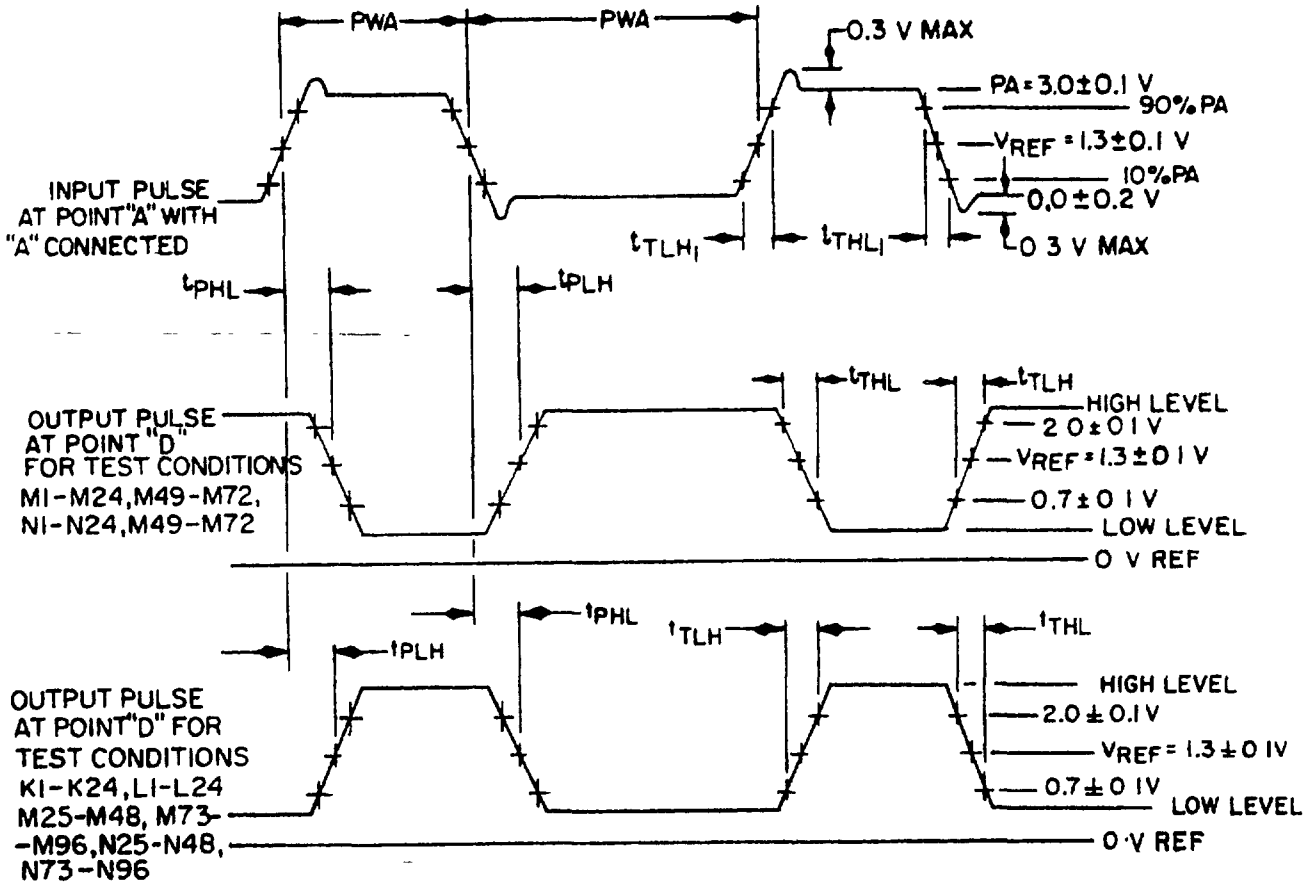
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N73	1C Propagation delay time (t_{PHL}), other input low, C_L = 140 pF	25	Open	4	26	10	180	ns	12, 13
N74	2C	27		6	5				
N75	3C	23		3	24				
N76	4C	21		2	22				
N77	5C	31		14	13				
N78	6C	32		12	33				
N79	7C	29		9	8				
N80	8C	30		28	7				
N81	9C	19		40	39				
N82	10C	37		18	38				
N83	11C	36		35	15				
N84	12C	16		17	34				
N85	1C	4		25	26				
N86	2C	6		27	5				
N87	3C	3		23	24				
N88	4C	2		21	22				
N89	5C	14		31	13				
N90	6C	12		32	33				
N91	7C	9		29	8				
N92	8C	28		30	7				
N93	9C	40		19	39				
N94	10C	18		37	38				
N95	11C	35		36	15				
N96	12C	17		16	34				

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* FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns
   PWA = 500 ±50 ns
2. Before making the measurements specified in tables XII, XIII, XIV, and XV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.
3. All voltages are referenced to ground.

* FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV and XV.

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TABLE XVI. Output crosstalk (low-level) test.

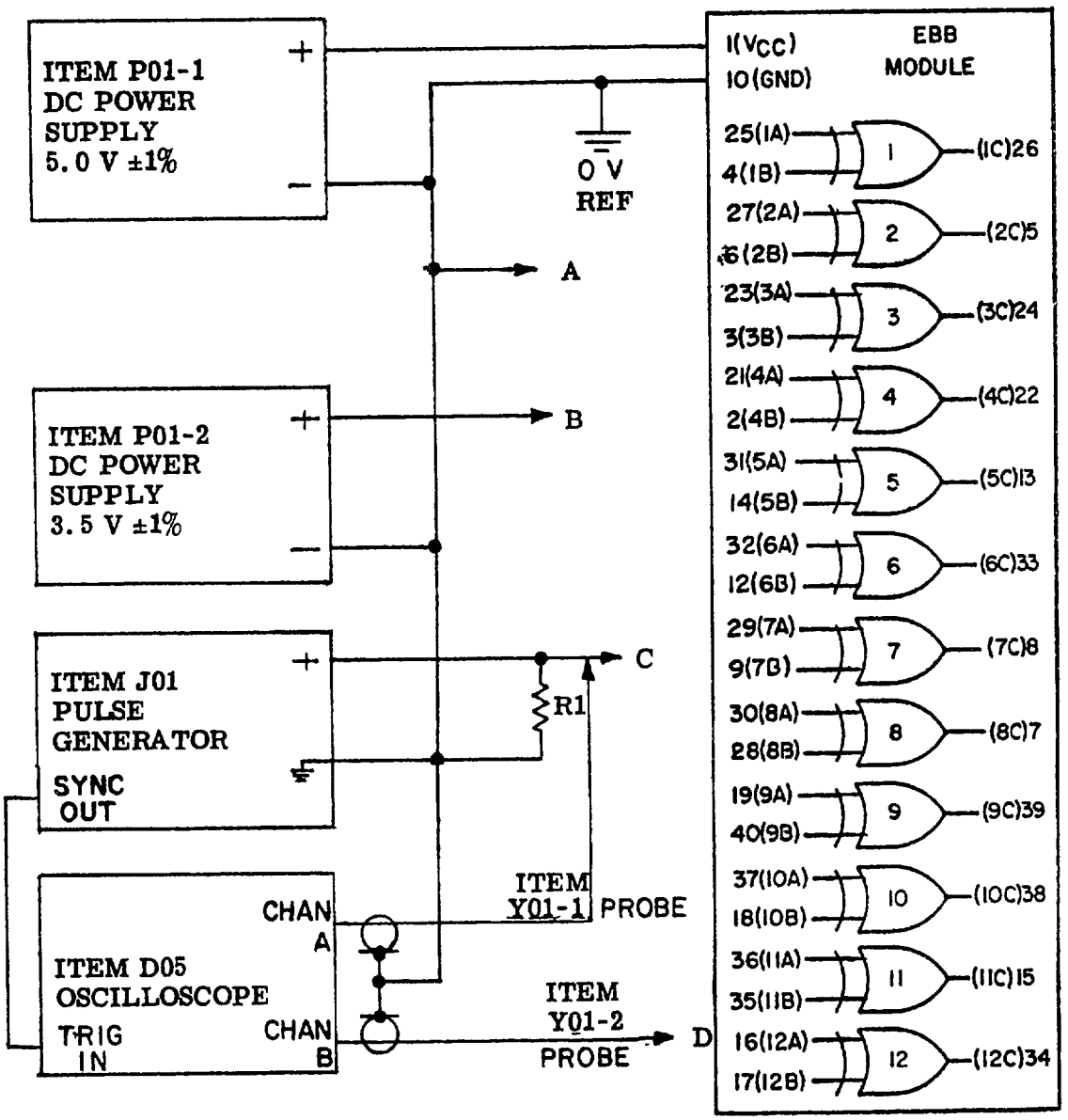
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1C Output crosstalk, low level	25, 4	Open	27,23,21,31, 32,29,30,19, 37,36,16	26	0	400	mV	14, 15
P2	2C	27, 6		25,23,21,31, 32,29,30,19, 37,36,16	5				
P3	3C	23, 3		25,27,21,31, 32,29,30,19, 37,36,16	24				
P4	4C	21, 2		25,27,23,31, 32,29,30,19, 37,36,16	22				
P5	5C	31, 14		25,27,23,21, 32,29,30,19, 37,36,16	13				
P6	6C	32, 12		25,27,23,21, 31,29,30,19, 37,36,16	33				
P7	7C	29, 9		25,27,23,21, 31,32,30,19, 37,36,16	8				
P8	8C	30, 28		25,27,23,21, 31,32,29,19, 37,36,16	7				
P9	9C	19, 40		25,27,23,21, 31,32,29,30, 37,36,16	39				
P10	10C	37, 18		25,27,23,21, 31,32,29,30, 19,36,16	38				
P11	11C	36, 35		25,27,23,21, 31,32,29,30, 19,37,16	15				
P12	12C	16, 17		25,27,23,21, 31,32,29,30, 19,37,36	34				

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1C Output crosstalk, high level	25	4	27,23,21,31,32, 29,30,19,37,36, 16	26	0	400	mV	14, 15
R2	2C	27	6	25,23,21,31,32, 29,30,19,37,36, 16	5				
R3	3C	23	3	25,27,21,31,32, 29,30,19,37,36, 16	24				
R4	4C	21	2	25,27,23,31,32, 29,30,19,37,36, 16	22				
R5	5C	31	14	25,27,23,21,32, 29,30,19,37,36, 16	13				
R6	6C	32	12	25,27,23,21,31, 29,30,19,37,36, 16	33				
R7	7C	29	9	25,27,23,21,31, 32,30,19,37,36, 16	8				
R8	8C	30	28	25,27,23,21,31, 32,29,19,37,36, 16	7				
R9	9C	19	40	25,27,23,21,31, 32,29,30,37,36, 16	39				
R10	10C	37	18	25,27,23,21,31, 32,29,30,19,36, 16	38				
R11	11C	36	35	25,27,23,21,31, 32,29,30,19,37, 16	15				
R12	12C	16	17	25,27,23,21,31, 32,29,30,19,37, 36	34				

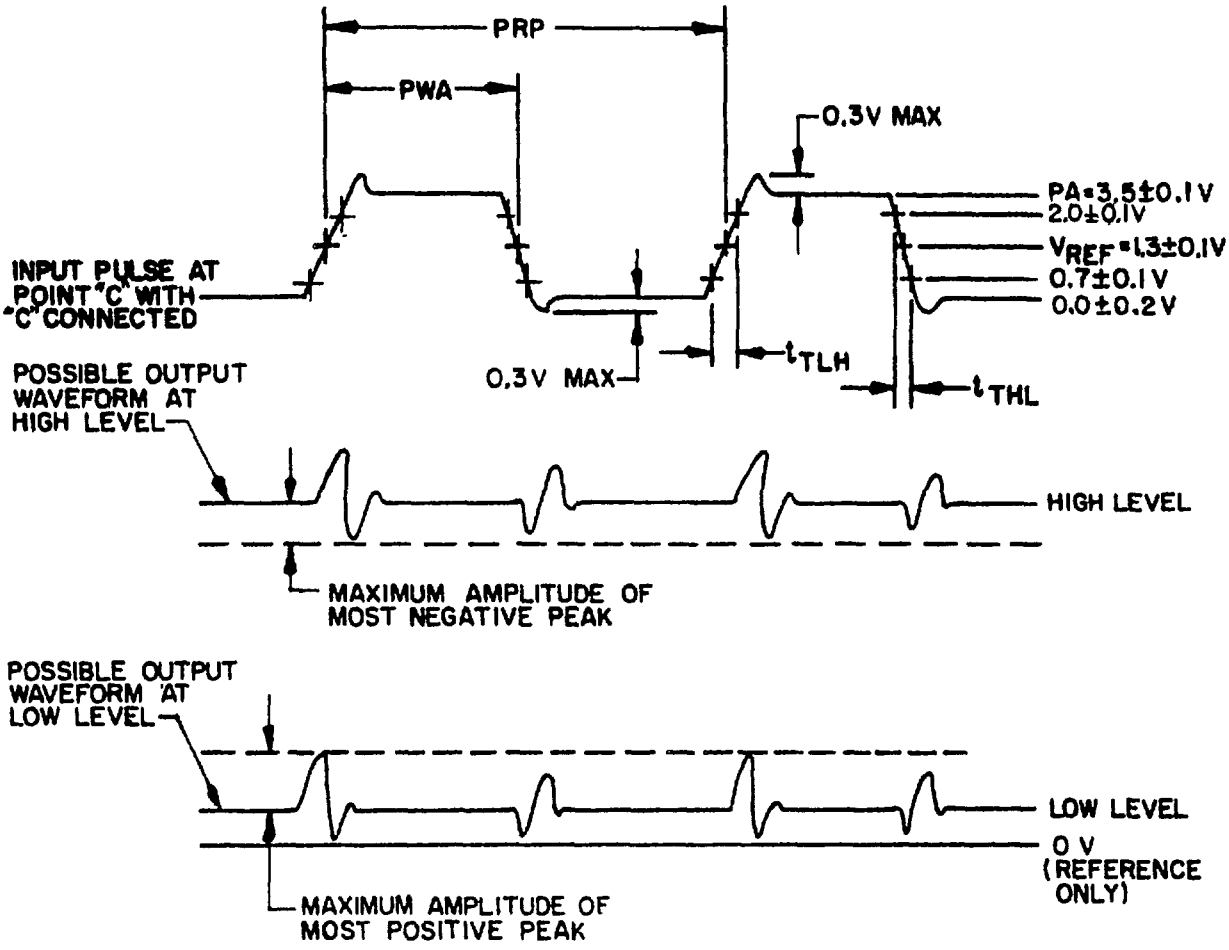
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load each output as specified on figure 9.

* FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at all inputs).
   PWA = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to ground.

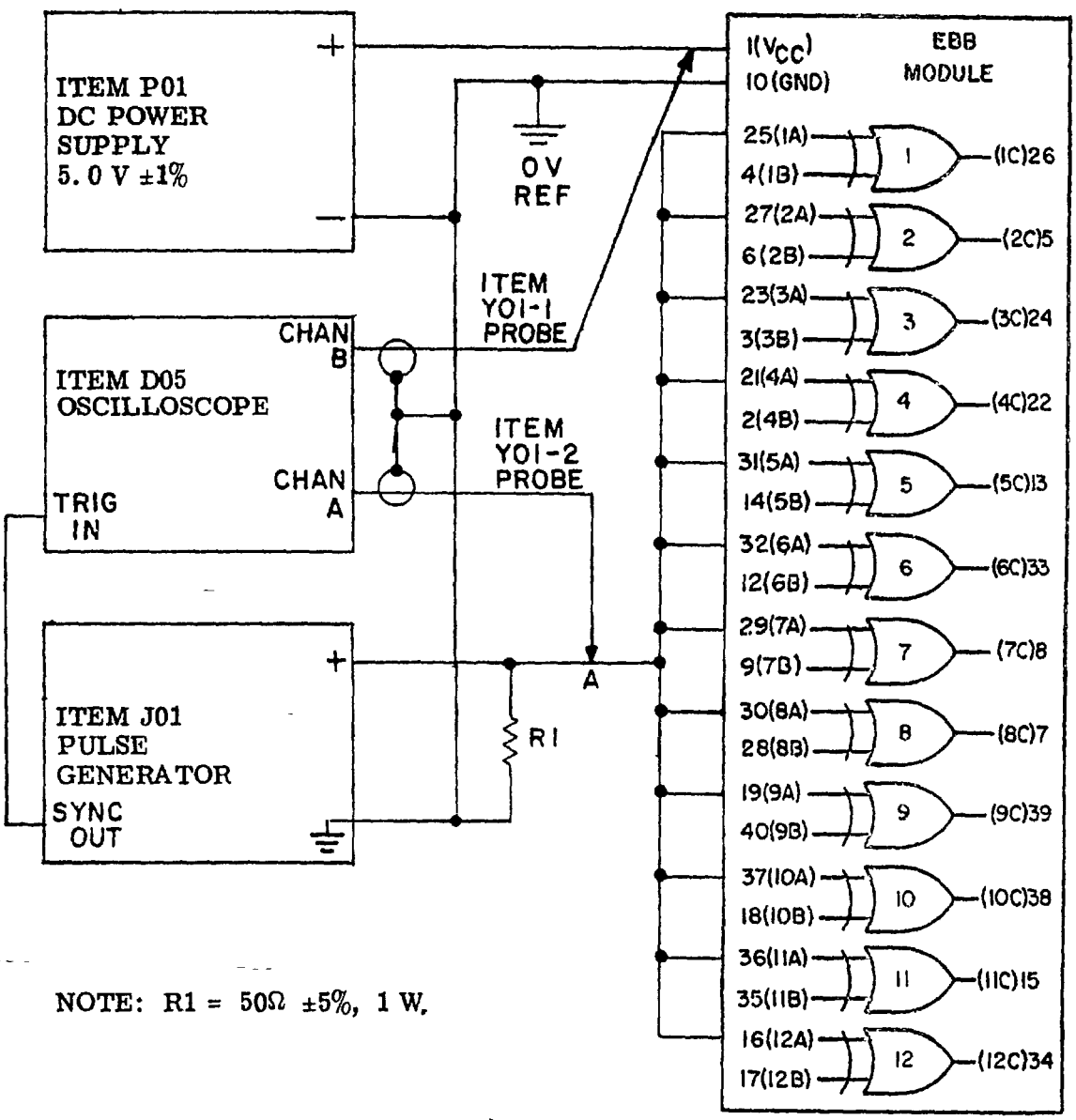
* FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	300	mV	16, 17

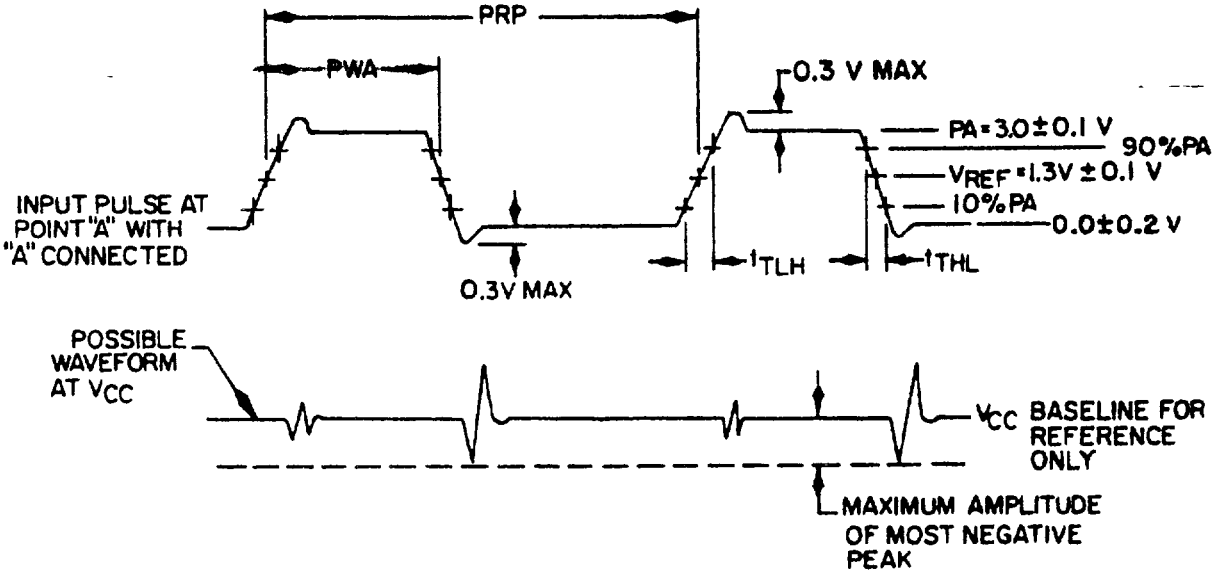
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NOTE: R1 = 50Ω ±5%, 1 W.

* FIGURE 16. Power supply transient amplitude test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 20 \pm 2$ ns.
   PWA = 200 ±20 ns.
   PRP = 1000 ±100 ns.
2. All voltages are referenced to ground.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

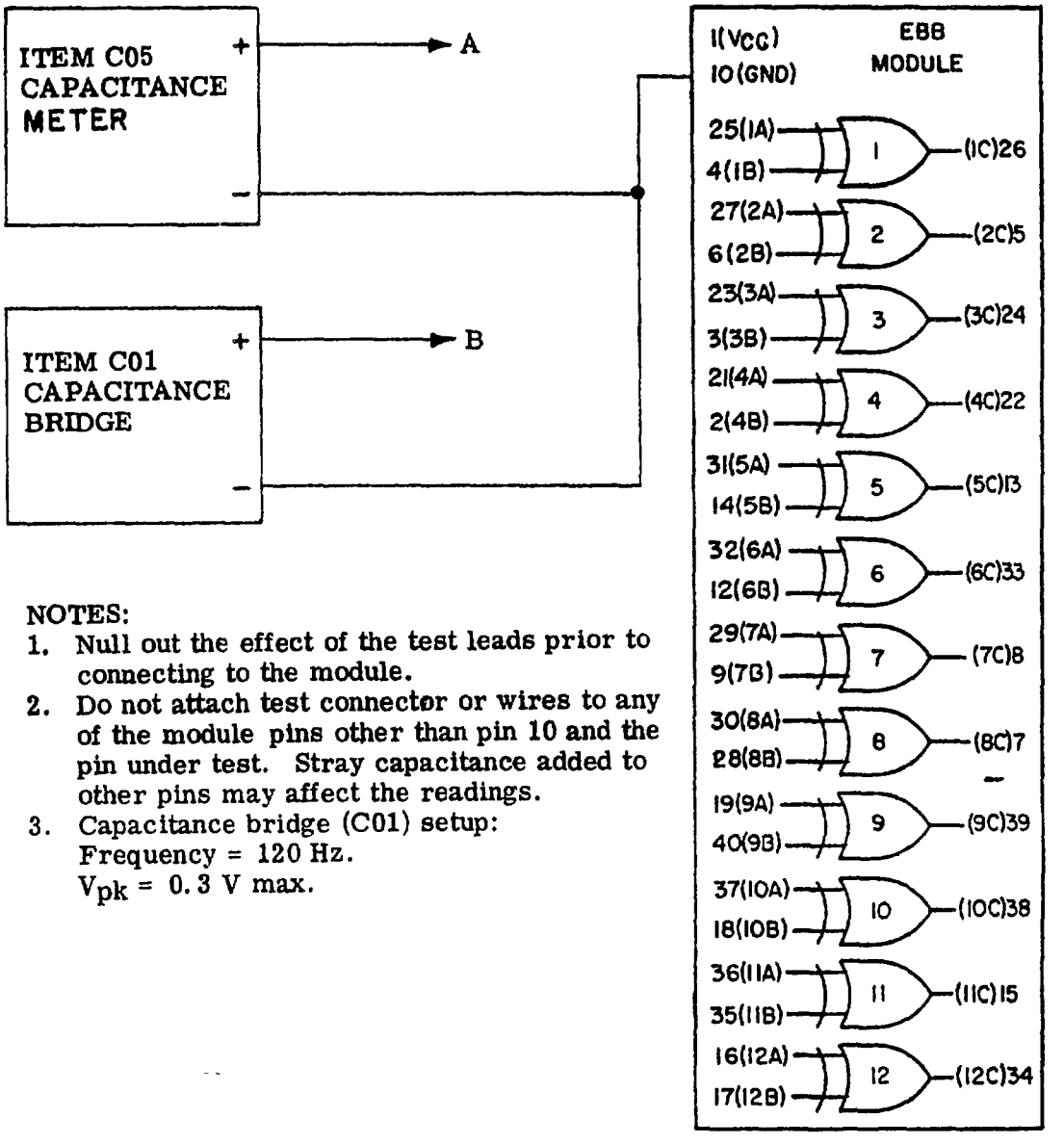
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	1.65	2.75	µF	18

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	1A Input capacitance	25	Open	2	15	pF	18
U2	1B	4					
U3	2A	27					
U4	2B	6					
U5	3A	23					
U6	3B	3					
U7	4A	21					
U8	4B	2					
U9	5A	31					
U10	5B	14					
U11	6A	32					
U12	6B	12					
U13	7A	29					
U14	7B	9					
U15	8A	30					
U16	8B	28					
U17	9A	19					
U18	9B	40					
U19	10A	37					
U20	10B	18					
U21	11A	36					
U22	11B	35					
U23	12A	16					
U24	12B	17					

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V max.

* FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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